

Deriving Relativistic Time Dilation from the Spatial Distribution of Mass: A Unified Origin of the Lorentz and Schwarzschild Metrics

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ABSTRACT: Inertial Relativity and its key formula the Space-Time Equivalence (STE)¹ originally hypothesized a universal mathematical relationship between time and the spatial distribution of matter: *The characteristic length scale factor that determines time dilation between any two physical systems about any respective axes is universally equal to the fifth root of the inverse ratio of their moments of inertia about the same axes.* A central prediction of this hypothesis is that General and Special Relativity are edge cases of the STE. In our prior paper², we began validating this prediction by arriving at General Relativity calculations with exact precision utilizing only principles of relativistic inertia. This paper extends the validation by demonstrating the mathematical equivalence between the STE, General Relativity, and Special Relativity. In the context of moment of inertia about an axis, we demonstrate that the STE is mathematically equivalent to Schwarzschild gravitational time dilation. In the context of linear inertia (mass), the STE states an inverse third root relationship with inertia (mass), which we demonstrate is mathematically equivalent to Lorentz time dilation. Numerical evaluation across the full parameter space of mass, radius, and density variation shows that the fifth-root inertia relation $k_i = (I_2/I_1)^{1/5}$ reproduces the time dilation ratio exactly under isometric scaling (equal density between the two systems) and requires generalization for the case of differing densities. The general form is the compactness ratio $k_d = (I_1/I_2)(R_2/R_1)^3 = D_1/D_2 = (M_1 R_2)/(M_2 R_1)$, equivalently expressible in (I, ρ) coordinates as $(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$. The fifth-root inertia form is the constant-density limit of k_d . The generalized STE confirms the central prediction: Special and General Relativity emerge as edge cases of the same underlying phenomenon, Inertial Relativity. Relativistic time dilation between any two systems follows from the spatial distribution of their mass.

Keywords: modern physics, inertial relativity, space-time equivalence, time dilation, general relativity, special relativity, moment of inertia, compactness, density, unification

1 Introduction

This paper presents a mathematical derivation connecting the Space-Time Equivalence (STE), General Relativity³, and Special Relativity⁴. In doing so, it validates the hypothesis introduced in “The Universe of Light”¹: that gravitational and velocity-induced time dilation are edge cases of a universal relationship between time and inertia. We build upon DeGerlia’s prior work introducing the concept of Inertial Density and predicting time dilation calculations with exact precision², which established that the Schwarzschild condition reduces to a constant threshold of mass over mean radius, the DeGerlia Threshold ($D_{crit} = 6.73295 \times 10^{26}$

kg/m), about the selected axis. From these demonstrations, the mathematical equivalence between the STE, General Relativity, and Special Relativity is established, unifying the two theories under a single framework: Inertial Relativity.

The original hypothesis, that time dilation between two systems is determined by the fifth root of their inverse moment-of-inertia ratio, is presented and developed in Sections 2–5. Numerical evaluation across the full parameter space of mass, radius, and density variation (Appendix Appendix B) shows that this relation is exact under isometric (constant-density) scaling and requires generalization for non-isometric system pairs. The generalized form (Equation 23) reduces to the originally hypothesized form in the constant-density limit. Section 7 presents the generalized framework and its consequences.

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2 Inertial Relativity

Inertial Relativity is a principle introduced by the author in a prior paper "The Universe of Light"¹. In that work, we hypothesize that General and Special Relativity emerge from Inertial Relativity. This paper demonstrates that equivalence.

Inertia is relativistic, meaning that without a comparator, any measurement of it would have no meaning. For example, something with a moment of inertia $1 \times 10^{20} \text{ kg m}^2$ might be perceived as difficult to rotate, but not more difficult than something with $1 \times 10^{30} \text{ kg m}^2$. However, the concept of relativity also says that how something is observed is influenced by the frame of observation. A system, in this context, is any collection of mass and spatial extent one chooses to define. It need not be spherical, uniform, high density, or even massive. A system may be a planet, a uniform sphere, a supermassive black hole, a disc galaxy, a cube, a cubic sample of a cumulus cloud, or a system consisting of a hydrogen proton and a single electron.

2.1 Rotational Inertia

Moment of inertia for a system is distinct about every axis in the system. For any given axis, the moment of inertia dictates how an applied torque translates into angular acceleration. Moment of inertia is defined as:

$$I = \sum_i m_i r_i^2 \quad (1)$$

where r_i is the perpendicular distance from each mass element m_i to the chosen axis. Just as mass governs how a system responds to linear force ($F = ma$), moment of inertia governs how it responds to torque ($\tau = I\alpha$).

2.2 Linear Inertia

Linear inertia is mass:

$$M = \sum_i m_i \quad (2)$$

where m_i are the constituent mass elements. Linear inertia defines how an applied force translates into translational acceleration. It is quantified entirely by mass ($F = ma$). It is dimensional, operating along a specific spatial vector, but unlike moment of inertia, it is isotropic: mass offers the same resistance to changes in motion regardless of the direction of the vector.

2.3 Isometric Scaling

Isometric scaling is the uniform scaling of a system in all spatial dimensions while maintaining constant density. Under isometric scaling, every length in the system scales by the same factor k :

$$\ell' = k\ell \quad (3)$$

where ℓ is the characteristic length of the system and k is the linear scale factor. Isometric scaling of a system by a factor of k results in the following changes to mass, radius, and moment of inertia:

$$M' = k^3 M \quad (4)$$

where M is the mass and k is the linear scale factor.

$$R' = kR \quad (5)$$

where R is the mean radius about the selected axis.

$$I' = k^5 I \quad (6)$$

where I is the moment of inertia about a selected axis; k is the linear scale factor.

2.4 Inertia Abstracts Mass and Radius

Because every change in length necessarily changes the system's mass, Equations 3–5 hold only for strict isometric scaling; Equation 6, however, abstracts mass and radius into moment of inertia. If two systems share moment of inertia about any axis, then an identical torque applied to each system about that same axis will produce identical changes in motion, regardless of the system geometry; independent of mass and radius individually. The moment of inertia alone dictates how a system interacts with time about a particular axis. From this, we can conclude the following formula is universal and is not bound in any way to isometric scaling. The derivations that follow demonstrate that General and Special Relativity emerge from this principle.

2.5 Universality

The extension from isometrically scaled systems to all systems follows from the nature of moment of inertia itself. Because I fully determines rotational response to applied torque, two systems with the same I are dynamically indistinguishable regardless of geometry. A thin shell and a solid sphere with the same I exhibit the same rotational physics.

The fifth root relationship therefore holds universally: for any two systems about any two respective axes, the characteristic length scale factor that determines time dilation is the fifth root of the inverse ratio of their moments of inertia about those same axes. Isometric scaling establishes the exponent (Equation 6).

2.6 The DeGerlia Time Dilation Framework

Inertial density P is the moment of inertia of a system divided by its volume:

$$P = I/V \quad (7)$$

Since linear inertia is mass, this reduces directly to ordinary density $\rho = M/V$ in the linear case; inertial density is the rotational generalization of a quantity already familiar from linear mechanics.

For a uniform sphere, substituting $I = \frac{2}{5}MR^2$ and $V = \frac{4}{3}\pi R^3$, this reduces to:

$$P = \frac{M}{R} \times \frac{3}{10\pi} \quad (8)$$

Just as ordinary density ($\rho = M/V$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion*. Because P is defined relative to a selected axis, it can be evaluated per axis directly rather than requiring a full tensor treatment

when the quantity varies by direction.

DeGerlia Compactness $D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over mean radius ratio for a system about the selected axis.

Setting $R = r_s$ (the Schwarzschild condition)⁵ and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia Compactness at which a system undergoes gravitational collapse:

$$D_{crit} = \frac{c^2}{2G} = 6.73295 \times 10^{26} \text{ kg/m} \quad (9)$$

The normalized DeGerlia Compactness D_{norm} is the ratio of a system's DeGerlia Compactness to the DeGerlia Threshold:

$$D_{norm} = \frac{D}{D_{crit}} \quad (10)$$

D_{norm} is algebraically identical to r_s/R and is therefore a linear scale factor k : the value that enters directly into $d\tau/dt = \sqrt{1-k}$. Because D_{crit} is a universal constant, D_{norm} places every system on the same scale, making D_{norm} values directly comparable across systems of any mass. The time dilation follows directly:

$$\frac{d\tau}{dt} = \sqrt{1 - D_{norm}} \quad (11)$$

Substituting $D = M/R$:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{M}{R \cdot D_{crit}}} \quad (12)$$

The **DeGerlia Time Dilation** (DTD) is the square root of the normalized compactness:

$$\text{DTD} = \sqrt{D_{norm}} = \sqrt{\frac{D}{D_{crit}}} \quad (13)$$

Since $D_{norm} = k$, the gravitational time dilation is:

$$\frac{d\tau}{dt} = \sqrt{1 - \text{DTD}^2} \quad (14)$$

and the two quantities satisfy:

$$\text{DTD}^2 + \left(\frac{d\tau}{dt}\right)^2 = 1 \quad (15)$$

For systems in weak gravitational fields, $d\tau/dt \approx 1 - k/2$; the time dilation signal is confined to digits far from the leading unity, limiting the effective precision of the representation. DTD expresses this signal directly. Because $\sqrt{x}$ is algebraically homogeneous while $\sqrt{1-x}$ is not, DTD ratios decompose into independent mass and mean radius contributions:

$$\frac{\text{DTD}_1}{\text{DTD}_2} = \sqrt{\frac{D_1}{D_2}} = \sqrt{\frac{M_1}{M_2}} \cdot \sqrt{\frac{R_2}{R_1}} \quad (16)$$

The gravitational time dilation ratio $\sqrt{1-k_1}/\sqrt{1-k_2}$ does not factor; the nonlinearity of $f(x) = \sqrt{1-x}$ precludes decomposition into constituent ratios. DTD is defined for all $D > 0$ and diverges as $D \rightarrow \infty$, correctly reflecting the unbounded growth of the dilation parameter at and beyond the collapse threshold, where the gravitational time dilation form $\sqrt{1-k}$ is undefined. DTD de-

pends solely on D/D_{crit} , requiring no system-specific reference values such as the Schwarzschild radius.

2.7 D_{crit} , D_{norm} , and the Role of Ratios

D_{crit} and D_{norm} play distinct roles in the framework, and the distinction matters for how precision propagates.

D_{crit} is a universal constant. It is the compactness at the collapse threshold, the same value for every system. Its numerical value derives from $c^2/(2G)$ and inherits the experimental precision of G — bounded at approximately six significant figures.

D_{norm} is system-specific. For a given system, $D_{norm} = D/D_{crit}$ is the fraction of the universal threshold that system has reached. Every system has its own D_{norm} , determined by its mass and mean radius.

When time dilation is computed for a single system through $d\tau/dt = \sqrt{1-D_{norm}}$ or $\text{DTD} = \sqrt{D_{norm}}$, the result inherits the precision of D_{crit} — and therefore of G . Single-system quantities cannot escape this precision floor.

When time dilation is computed as a comparison between two systems, D_{crit} appears identically in both halves of the ratio and cancels structurally:

$$\frac{D_{norm,1}}{D_{norm,2}} = \frac{D_1/D_{crit}}{D_2/D_{crit}} = \frac{D_1}{D_2}$$

The result no longer depends on D_{crit} at all. Comparative predictions are decoupled from the precision of G and bounded only by the precision of the mass and length inputs, which can be measured to many more digits than G .

The same property holds for any universal constant. G , c , and any dimensional combination of universal constants appear identically on both sides of any ratio between two systems, and cancel. The framework's comparative results are therefore robust even to the hypothetical scenario in which G and c are scale-dependent rather than strictly constant: as long as they vary together such that their combinations (and therefore D_{crit}) remain identical between the two systems being compared, the cancellation still holds.

This is the structural reason the framework prioritizes ratios. Time dilation is a relational property — the natural object is the comparison between two systems, not the absolute time dilation of one. Single-system quantities require a universal reference to express and inherit that reference's precision; comparative quantities do not. The relational form is both more physically meaningful (no fictional flat-spacetime observer at infinity is needed) and more numerically precise (no inherited precision floor from universal constants).

3 The Space-Time Equivalence (STE)

The STE is the core universal equivalence stated by Inertial Relativity. In the original work¹, the author hypothesized:

“Ultimately, general relativity, gravitational, and velocity-induced time dilation are phenomena associated with the law of Space-Time Equivalence. They are static-scale frames of reference (from the human perspective), extreme scenarios of velocity or gravity, and

the time dilation described by Einstein—these are likely edge cases of Space-Time Equivalence.”

The universal scale factor between any two systems is the compactness ratio:

$$k_d = \frac{I_1}{I_2} \cdot \left(\frac{R_2}{R_1} \right)^3 = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \quad (17)$$

where I_1, I_2 are the moments of inertia, R_1, R_2 are the mean radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGleria compactness of system i . By convention, System 1 denotes the observer and System 2 the observed, consistent with the Schwarzschild convention. The three forms in Equation 17 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 exposes the compactness-ratio structure, and $M_1 R_2/(M_2 R_1)$ is the fully expanded form.

The DTD ratio between two systems follows directly from k_d :

$$\frac{\text{DTD}_1}{\text{DTD}_2} = \sqrt{k_d} \quad (18)$$

Equation 18 is the universal STE. The single-system Pythagorean relation $\text{DTD}^2 + (d\tau/dt)^2 = 1$ (Equation 15) provides the complementary GTD form for any individual system; the relationship between DTD and GTD as alternative computational representations is developed in Section 7.

Under isometric scaling (constant density), the isometric equivalence holds:

$$k_i = \left(\frac{I_2}{I_1} \right)^{1/5} \quad (19)$$

where k_i is the linear scale factor derived from the inertia ratio. Under isometric conditions the two systems share the same density, and the fifth root of the inertia ratio recovers the linear scale factor exactly. Subsequent references to this specific inertia-derived form use k_i .

In the linear context, inertia is M and the linear inertia version of the isometric equivalence applies (see Appendix A). Special Relativity describes linear inertia:

$$k_m = \left(\frac{M_2}{M_1} \right)^{1/3} \quad (20)$$

where M_1 and M_2 are the masses of the two systems and k_m is the linear scale factor derived from the mass ratio. Subsequent references to this specific mass-derived form use k_m .

3.1 Relating Linear Scale Factor k to General Relativity

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \quad (21)$$

where $d\tau/dt$ is the ratio of proper time to coordinate time and k is the linear scale factor derived from inertia. Specific forms include k_i (Equation 19, isometric scaling) and k_m (Equation 20, static-radius linear case). Every expression of time dilation takes this form; k is always a linear scale factor derived from the relative inertia of two systems about their respective selected axes.

3.2 Properties of Time

The STE reveals the following properties of time:

- Time is relativistic: it has meaning only when compared, and how it appears depends on the frame of observation
- Time emerges from inertia
- Time is dimensional in three spatial dimensions
- Time is multimodal: both linear and rotational
- Time is dynamic for every system or subsystem
- Time dilation applies to all systems, not only uniform or spherically symmetric ones
- Time is universally and precisely determined by the spatial distribution of matter in the system, as captured by inertia, about and along every axis, whether or not in motion or being observed
- Time dilation is system-relational: no observer experiences their own time as anything but normal; dilation is a structural property of the comparison between two systems

3.3 Physical Interpretation

The STE states that the relative pace of time for any system about any axis is determined by its moment of inertia about that same axis. The more inertia a system has about that axis, the slower its clock runs relative to a system with less inertia. This is consistent with the notion that time is a measure of change, and inertia resists change. Therefore, the more inertia a system has, the less it changes over a given period of time, and thus the slower its system clock runs relative to a system with less inertia.

At any given time, any system is always undergoing some degree of rotation and some amount of linear motion, however small. This law applies to any system whose mass is distributed over some region of space, requiring, at minimum, both mass and spatial extent. One may define the system as the core of a planet, a disc-shaped galaxy, a cosmic filament, a spherical sample of interstellar space, or even an asymmetric black hole described by the Kerr metric⁶—a topic for future investigation.

3.4 STE Example

Consider a uniform sphere of radius 1 m and mass 1 kg scaled isometrically by a factor of two, producing a second sphere of radius 2 m and mass $2^3 = 8$ kg. Their moments of inertia are:

$$I_1 = \frac{2}{5}(8\text{ kg})(2\text{ m})^2 = 12.8\text{ kg m}^2 \quad I_2 = \frac{2}{5}(1\text{ kg})(1\text{ m})^2 = 0.4\text{ kg m}^2$$

The ratio $I_1/I_2 = 32 = 2^5$ confirms that moment of inertia scales as k_i^5 under isometric scaling. Designating the larger sphere as the observer (System 1) and the smaller as the observed (System 2), the isometric equivalence yields:

$$k_i = \left(\frac{I_2}{I_1} \right)^{1/5} = \left(\frac{0.4}{12.8} \right)^{1/5} = \left(\frac{1}{32} \right)^{1/5} = 0.5$$

recovering the linear scale factor between the two systems. The resulting time dilation is:

$$\frac{d\tau}{dt} = \sqrt{1 - k_i} = \sqrt{1 - 0.5} \approx 0.707$$

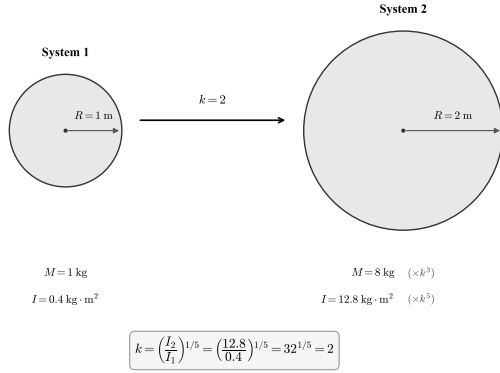


Fig. 1 Two uniform spheres related by isometric scaling. The isometric equivalence recovers the linear scale factor $k_i = 0.5$ from the moment of inertia ratio; time dilation follows directly.

3.5 Density-Explicit Form

The STE (Equation 18) can be expressed equivalently in terms of density. For a uniform body $I = \frac{2}{3}MR^2$ with $M = \frac{4}{3}\pi R^3\rho$, so $I \propto \rho R^5$ and:

$$\left(\frac{I_1}{I_2}\right)^{1/5} = \left(\frac{\rho_1}{\rho_2}\right)^{1/5} \cdot \frac{R_1}{R_2} \quad (22)$$

Substituting into the STE:

$$\frac{DTD_1}{DTD_2} = \left(\frac{I_1}{I_2}\right)^{1/5} \cdot \left(\frac{\rho_1}{\rho_2}\right)^{3/10} \quad (23)$$

The $3/10$ exponent ($= 1/2 - 1/5$) on the density ratio arises from the difference between the density dependence in the DTD ratio ($1/2$ power) and the inertia ratio ($1/5$ power). Under constant density this factor is unity and k_i (Equation 19) gives the DTD ratio directly. Equation 23 is the universal STE; k_i is its constant-density projection. The pair comparison tables (Appendix Appendix B) demonstrate the universal form's necessity in the non-isometric cases, where the density correction is non-unity.

3.6 Comparison of Relative Time Paces

The universal scale factor k_d (Equation 17) is independent of D_{crit} : dividing each system's normalized compactness D/D_{crit} cancels the threshold, leaving a direct ratio of compactnesses. This is the *raw* scale factor. When the denominator system is the Schwarzschild condition ($D_2 = D_{crit}$), k_d reduces to the *normalized* linear scale factor:

$$k = \frac{D}{D_{crit}} = D_{norm}$$

equivalent to r_s/R —the compactness parameter of general relativity. The Schwarzschild case is the specific instance of compar-

ing a system to its own collapse point—same mass, physical mean radius R versus Schwarzschild radius r_s .

Applying Equation 21, $d\tau/dt = \sqrt{1 - k}$, converts a normalized linear scale factor into time dilation. The value of k must be normalized ($k = D_{norm}$) for $\sqrt{1 - k}$ to yield gravitational time dilation: the pace of time relative to flat spacetime.

Property: The ratio of two normalized values is always raw.

$$\frac{D_{norm,1}}{D_{norm,2}} = \frac{D_1/D_{crit}}{D_2/D_{crit}} = \frac{D_1}{D_2}$$

D_{crit} cancels, leaving a direct ratio of compactnesses with no reference to the collapse threshold. A raw linear scale factor cannot be passed into $\sqrt{1 - k}$ to obtain a time dilation; it must first be normalized by dividing by D_{crit} .

Each system has its own time dilation $d\tau/dt = \sqrt{1 - k}$, where k is that system's normalized linear scale factor. The relative pace of time between two systems is the ratio of their time dilations:

$$\frac{\sqrt{1 - k_2}}{\sqrt{1 - k_1}}$$

This is not equal to $\sqrt{1 - k_1/k_2}$, because k_1/k_2 is raw— D_{crit} has cancelled—and cannot be substituted into $\sqrt{1 - k}$ without renormalization. To recover the linear scale factor from a known time dilation:

$$k = 1 - \left(\frac{d\tau}{dt}\right)^2$$

The ratio of the recovered k values yields the raw compactness ratio D_1/D_2 .

Example (Comparison of Relative Time Paces): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^7$ m. S_2 : $M_2 = 1 \times 10^{20}$ kg, $R_2 = 2 \times 10^{-7}$ m.

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{(1 \times 10^{30})(2 \times 10^{-7})}{(1 \times 10^{20})(1 \times 10^7)} = 2.00000 \times 10^{-4}$$

Time dilation of S_2 :

$$D_2 = \frac{M_2}{R_2} = \frac{1 \times 10^{20}}{2 \times 10^{-7}} = 5.00000 \times 10^{26} \text{ kg/m}$$

$$\frac{D_2}{D_{crit}} = \frac{5.00000 \times 10^{26}}{6.73295 \times 10^{26}} = 7.42617 \times 10^{-1}$$

$$\frac{d\tau}{dt} = \sqrt{1 - 7.42617 \times 10^{-1}} = 5.07330 \times 10^{-1}$$

Confirmation: Schwarzschild gravitational time dilation:

$$k_2 = \frac{2GM_2}{R_2 c^2} = \frac{2(6.67430 \times 10^{-11})(1 \times 10^{20})}{(2 \times 10^{-7})(8.98755 \times 10^{16})} = 7.42617 \times 10^{-1}$$

$$\frac{d\tau}{dt} = \sqrt{1 - k_2} = \sqrt{1 - 7.42617 \times 10^{-1}} = 5.07330 \times 10^{-1}$$

Relative time dilation between S_1 and S_2 :

$$k_1 = \frac{2GM_1}{R_1 c^2} = \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^7)(8.98755 \times 10^{16})} = 1.48523 \times 10^{-4}$$

$$\frac{\sqrt{1-k_2}}{\sqrt{1-k_1}} = \frac{5.07330 \times 10^{-1}}{9.99926 \times 10^{-1}} = 5.07368 \times 10^{-1}$$

4 General and Special Relativity

Schwarzschild gravitational time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{R}} \quad (24)$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius and R is the mean radius about the selected axis.

Lorentz time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}} \quad (25)$$

where v is the velocity of the moving frame and c is the speed of light.

These are the same form as Equation 21, where $k = r_s/R$ in the gravitational case and $k = v^2/c^2$ in the velocity case. The two expressions are equivalent: the escape velocity at mean radius R is $v^2 = 2GM/R$, so $v^2/c^2 = 2GM/(Rc^2) = r_s/R$.

Schwarzschild gravitational time dilation is the STE (Equation 17) evaluated under the constraint $M_1 = M_2$ (same mass, different radii). Lorentz time dilation is the STE evaluated under the constraint $R_1 = R_2$ (same radius, different masses). Both are projections of the same compactness-based relation; the derivations of each projection are developed in Section 5.

5 Expressions of Relative Inertial Linear Scale

The parameter k is always the linear scale factor between two systems about their respective selected axes. Because k is determined by relative inertia, it is fundamentally an inertial ratio. The various scenarios, such as special relativity, gravitational time dilation, etc., are all directly derived from this inertia ratio. In the case of general relativity, we simply set M_1 and M_2 equal and set R_2 to the Schwarzschild radius (see Appendix A).

Each expression of k below derives from k_i (Equation 19), where I_1 and I_2 are the moments of inertia of two systems about their respective selected axes. Substituting the moment of inertia for a uniform sphere, $I = \frac{2}{5}MR^2$:

$$k = \left(\frac{\frac{2}{5}M_2R_2^2}{\frac{2}{5}M_1R_1^2} \right)^{1/5} \quad (26)$$

where M_1, M_2 are the masses and R_1, R_2 are the mean radii of each system about the selected axis. The geometric constant $\frac{2}{5}$ cancels:

$$k = \left(\frac{M_2R_2^2}{M_1R_1^2} \right)^{1/5} \quad (27)$$

From here, each derivation applies specific constraints and solves.

5.1 Static Mass (Gravitational)

From Equation 27, setting $M_1 = M_2 = M$ and $R_2 = r_s$:

$$k = \left(\frac{Mr_s^2}{M \cdot R_1^2} \right)^{1/5} = \left(\frac{r_s^2}{R^2} \right)^{1/5} \quad (28)$$

where r_s is the Schwarzschild radius and R is the mean radius of the system about the selected axis. When mass cancels, the degree of homogeneity of the remaining quantity (R^2) is $L = 2$, so the exponent reduces from $1/5$ to $1/2$ (see Appendix A):

$$k = \left(\frac{r_s^2}{R^2} \right)^{1/2} = \frac{r_s}{R} \quad (29)$$

This is equivalently expressed through inertial density or DeGerlia Compactness, since $P \propto M/R$ and $D = M/R$. When mass is static, the ratio of either quantity reduces to the ratio of mean radii about the selected axis:

$$k = \frac{P_1}{P_2} = \frac{D_1}{D_2} = \frac{R_2}{R_1} = \frac{r_s}{R} \quad (30)$$

When compared against the DeGerlia Threshold $D_{crit} = 6.73295 \times 10^{26}$ kg/m:

$$k = \frac{D}{D_{crit}} \quad (31)$$

5.2 Static Radius (Linear)

From Equation 27, when $R_1 = R_2$, k_i reduces to (see Appendix A):

$$k_m = \left(\frac{M_2}{M_1} \right)^{1/3} \quad (32)$$

This is Equation 20. The Lorentz factor (Equation 25) has the same form as Equation 21, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling. Gravitational time dilation confirms this directly: $v^2/c^2 = r_s/R$ via the escape velocity identity, so the velocity-based k of special relativity and the radius-based k of general relativity are the same length ratio.

5.3 Constant Density (Isometric)

When density is held constant, $M \propto R^3$, and the isometric equivalence reduces to a pure linear scale ratio (see Appendix A):

$$k = \left(\frac{R_2^3 R_2^2}{R_1^3 R_1^2} \right)^{1/5} = \left(\frac{R_2^5}{R_1^5} \right)^{1/5} = \frac{R_2}{R_1} \quad (33)$$

5.4 General Case

When both mass and radius differ between two systems, no component cancels:

$$k_i = \left(\frac{I_2}{I_1} \right)^{1/5} = \left(\frac{M_2 R_2^2}{M_1 R_1^2} \right)^{1/5} \quad (34)$$

This is k_i (Equation 19) without constraint on mass or radius. The static-mass and static-radius cases derived above, producing Schwarzschild and Lorentz time dilation respectively, are edge

cases of k_i , each arising when one component of inertia is held constant. When densities differ between the two systems, k_i alone is insufficient and the universal STE (Equation 23) applies, equivalently expressible as $k_d = D_1/D_2$ (Equation 17).

5.5 Inertia–Time Dilation Identities

The preceding cases derive time dilation from inertia. The relationship is invertible: given the DTD ratio and the mean radius ratio, the moment of inertia ratio is fully determined. From Equation 16:

$$\frac{I_1}{I_2} = \left(\frac{DTD_1}{DTD_2} \right)^2 \cdot \left(\frac{R_1}{R_2} \right)^3 \quad (35)$$

The inertia ratio decomposes into a time dilation factor (the compactness ratio) and a spatial scale factor (the volume ratio). This is the STE (Equation 18) inverted: the $(R_1/R_2)^3$ factor converts the R^{-1} dependence in compactness back to the R^2 dependence in inertia.

Each system's DTD can also be computed directly from its moment of inertia. Substituting $M = 5I/(2R^2)$ into $D_{norm} = M/(R \cdot D_{crit})$:

$$DTD = \sqrt{\frac{5I}{2D_{crit}R^3}} \quad (36)$$

This expresses time dilation purely in terms of moment of inertia, geometry, and the DeGerlia Threshold, with no intermediate computation of mass or compactness.

6 Unification of GR and SR

From Equation 21, time dilation is always $d\tau/dt = \sqrt{1-k}$, and k is always the linear scale factor between two systems. In all cases, k reduces to a ratio of characteristic lengths. The Schwarzschild and Lorentz results are not independent discoveries; they are the same expression evaluated under different physical constraints on the same underlying parameter k .

As derived in Sections 5.1–5.5, each physical constraint yields a specific expression for k : static mass produces r_s/R , static radius produces $(M_2/M_1)^{1/3}$, constant density produces R_2/R_1 , and the unconstrained general case applies the STE (Equation 18). In every case, k reduces to a ratio of characteristic lengths. The velocity-based k of special relativity and the radius-based k of general relativity are connected directly through the escape velocity identity ($v^2/c^2 = r_s/R$). Both classical relativistic results emerge from this single expression without additional parameters, geometric similarity, or domain restrictions.

7 Analysis and Conclusion

The Space-Time Equivalence (Equation 21) states that time dilation is:

$$\frac{d\tau}{dt} = \sqrt{1-k}$$

where k is the linear scale factor derived from the relative compactness of two systems. Substituting the gravitational case ($k = r_s/R$) directly produces Schwarzschild time dilation (Equation 24). Substituting the velocity case ($k = v^2/c^2$, related to the gravitational case by the escape-velocity identity $v^2/c^2 = 2GM/(Rc^2) = r_s/R$) directly produces Lorentz time dilation (Equa-

tion 25). Equivalently, $k = D/D_{crit}$ gives the fraction of the DeGerlia Threshold consumed by the system, and is algebraically identical to r_s/R . Both Schwarzschild and Lorentz time dilation are produced exactly by the STE, with no free parameters.

7.1 Generalization of the Original Hypothesis

The original hypothesis stated that time dilation is determined by the fifth-root inertia relation $k_i = (I_2/I_1)^{1/5}$. Evaluation against the pair comparison data (Appendix Appendix B) confirms this relation under isometric (constant-density) scaling. For non-isometric pairs, the relation requires generalization. The generalized form is the compactness ratio:

$$k_d = \frac{I_1}{I_2} \cdot \left(\frac{R_2}{R_1} \right)^3 = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1}$$

equivalently expressible in (I, ρ) coordinates as:

$$\frac{DTD_1}{DTD_2} = \left(\frac{I_1}{I_2} \right)^{1/5} \cdot \left(\frac{\rho_1}{\rho_2} \right)^{3/10}$$

(Equation 23). These are two equivalent expressions of the generalized STE. The compactness-ratio form k_d exposes the inertial-geometric structure of the relationship directly; the (I, ρ) form makes explicit that compactness has two independent inputs — inertial scale and density. The fifth-root inertia form is recovered as the constant-density limit of either expression. Given the masses and mean radii of any two systems, the full relative time dilation between them follows from k_d .

7.2 Time Dilation Is Geometric, Not Gravitational

Compactness $D = M/R$ is a geometric quantity: mass distributed over a length. The label “gravitational time dilation” attaches historically because the effect was first observed in gravitational contexts, but no field-theoretic gravity entered the derivations above. Gravity is the name given to the observable consequence of compactness; the underlying cause is the relative geometric-inertial structure of the systems being compared. Time dilation is a structural property of how matter is distributed, not a property mediated by a force.

7.3 Time Dilation Is System-Relational, Not Personal

No observer experiences their own time as anything but normal. Time dilation arises only when the temporal property of one inertial system is compared against another. It is a structural property of the relationship between systems, not of any single system in isolation. Nothing physically changes for any observer on their own terms; what is structured is how two systems compare. This is the framing that the STE makes natural and that the standard treatment, by privileging the observer's frame, often obscures.

7.4 Special and General Relativity Are Projections of One Law

The Schwarzschild time dilation formula is the STE evaluated under the constraint $M_1 = M_2$: same mass, different radii. The Lorentz time dilation formula is the STE evaluated under the con-

straint $R_1 = R_2$: same radius, different masses, with the velocity-based form connected to the radius-based form by the escape-velocity identity. Neither is a separate theory. Both are evaluations of the same compactness-based relation under different choices of which variable is held fixed. The general case — both mass and radius varying, both densities varying — requires the universal STE (Equation 23). Special and General Relativity are not independent discoveries; they are two projections of a single law into the regimes where one or the other variable happens to be constrained.

7.5 G Cancels in Comparisons

The gravitational constant G appears identically in both halves of any time dilation ratio between two systems and therefore cancels structurally. The precision of comparative predictions in this framework is not bounded by the experimental uncertainty in G (~ 6 significant figures); it is bounded only by the precision of the mass and length inputs. Whatever the true value of G — including any joint scale-dependence with c that preserves the combination G/c^2 — comparative time dilation between two systems is exact at the precision of those inputs. The framework’s comparative predictions are decoupled from the precision floor of G .

7.6 DTD and GTD: Complementary Forms with Computational Advantages

The gravitational time dilation factor $d\tau/dt = \sqrt{1 - D/D_{crit}}$ answers “how much slower is this system’s clock relative to flat spacetime?” The DeGleria time dilation $DTD = \sqrt{D/D_{crit}}$ answers “how compact is this system relative to its collapse threshold?” The two are algebraically related ($GTD^2 + DTD^2 = 1$, Equation 15) and describe the same physical time dilation from different angles.

DTD is the natural computational form. The standard $\sqrt{1 - r_s/R}$ buries the time dilation signal under unity, with the meaningful information confined to digits far from the leading 1; DTD expresses the same physics directly, with the signal in the leading digits. DTD ratios also decompose cleanly:

$$\frac{DTD_1}{DTD_2} = \sqrt{\frac{M_1}{M_2}} \cdot \sqrt{\frac{R_2}{R_1}}$$

into independent mass and mean-radius contributions, a decomposition the $\sqrt{1 - k}$ form does not admit. Calculations may be carried in DTD throughout and converted back to standard $d\tau/dt$ at the end via $\sqrt{1 - DTD^2} = d\tau/dt$ when needed. Having both representations available makes the framework more flexible than the standard treatment.

7.7 The Schwarzschild Radius Is Not the Singularity

Standard treatments often conflate the Schwarzschild radius with the singularity, treating both as locations where physics breaks down. This framework keeps them properly distinct.

The *singularity* is the asymptote at $R \rightarrow 0$: the formal limit where compactness $D = M/R$ diverges to infinity. It is approached but never reached, and is the only point in the formulation where the framework itself is undefined.

The *Schwarzschild radius* is the point where $D = D_{crit}$, equivalently where the escape velocity equals c . It is a meaningful threshold — a milestone in the system’s compactness — but it is not a place where the framework fails. The standard $\sqrt{1 - r_s/R}$ form becomes imaginary at $R = r_s$ and forces a conflation between the threshold and the singularity. The DTD formulation remains real and well-defined throughout, including for $R < r_s$. The inertia–time dilation identities (Equations 35–36) continue to hold past the event horizon, depending only on ratios of real-valued quantities. The framework therefore gives a coherent description of the region between the Schwarzschild radius and $R \rightarrow 0$ — the region that standard general relativity treats as undefined or ambiguous.

7.8 Causality from Inertial Additivity

Both rotational and linear inertia are strictly additive (Equations 1 and 2). The temporal properties of any system are therefore wholly determined by its constituent matter, establishing that causality in this framework propagates exclusively from component to system. No top-down field assumption is required; the law operates at the level of mass distribution itself.

7.9 Future Extensions

The symmetric cases treated here are a subset of the framework’s domain. The (I, ρ) parameterization extends to arbitrary mass distributions, since moment of inertia is defined for any geometry. The D_{crit} derivation and Schwarzschild correspondence assume spherical symmetry and would need to be redone for asymmetric systems. Rotating systems described by the Kerr metric⁶ and non-spherical mass distributions are natural extensions enabled by the framework but not derived in this paper. The framework’s reach into weakly-gravitating and non-gravitational regimes follows from the same formulas returning vanishingly small but well-defined values for low compactness; the structural law does not break down. Cosmological implications follow if the framework extends consistently to light propagation, but those are the subject of separate work.

Appendix A Special-Case Derivations

The main text presents five expressions of k under different physical conditions (Sections 5.1–5.5). When a component of inertia is held constant between two systems, the exponent of the isometric equivalence changes from 1/5 to reflect the degree of homogeneity of the remaining quantity. This appendix derives each case from the general form using Euler’s theorem of homogeneous functions⁸, then verifies the general case through decomposition and numerical example.

The fundamental form of the isometric equivalence is $k = (I_2/I_1)^{1/L}$, where L is the degree of homogeneity of the compared quantity in the characteristic length. Under isometric scaling by factor k , a quantity with degree of homogeneity L scales as k^L . The specific exponents used throughout this paper (1/5 for moment of inertia, 1/3 for mass, 1/2 for R^2) each emerge from the value of L for that quantity.

$$k = \left(\frac{I_2}{I_1} \right)^{1/L} = \left(\frac{M_2 R_2^2}{M_1 R_1^2} \right)^{1/L} \quad (37)$$

Table 1 Isometric Scaling and Static Density: Scale factors for various properties under isometric scaling with constant density

Ratio	Metric	L
$M_a R_a^2 / M_b R_b^2$	Moment of inertia ($= MR^2$)	5
$I_{r,a} / I_{r,b}$	Moment of inertia	5
$I_{l,a} / I_{l,b}$	Linear inertia ($= M$)	3
M_a / M_b	Mass	3
V_a / V_b	Volume	3
R_a^2 / R_b^2	Radius squared	2
A_a / A_b	Area	2
R_a / R_b	Radius	1
ℓ_a / ℓ_b	Length	1

$I \propto MR^2$ where R is the mean radius about the selected axis.

Case 1: Static Mass

When $M_1 = M_2$ and $R_1 > R_2$:

$$k = \underbrace{\left(\frac{I_2}{I_1} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{MR_2^2}{MR_1^2} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{R_2^2}{R_1^2} \right)^{1/L}}_{L=2}$$

$$k = \left(\frac{R_2^2}{R_1^2} \right)^{1/2} = \frac{R_2}{R_1} \quad (38)$$

Example (Case 1): Gravitational time dilation compares a system at a given mass and mean radius to that same mass at the Schwarzschild radius, where time dilation is zero. This is a static-mass comparison between two uniform spheres. S_1 : $M = 1.9885 \times 10^{30}$ kg, $R_1 = 6.9570 \times 10^8$ m. S_2 : same mass at $R_2 = r_s = 2953.38$ m.

$$k = \frac{r_s}{R_1} = \frac{2953.38}{6.9570 \times 10^8} = 4.24520 \times 10^{-6}$$

$$\frac{d\tau}{dt} = \sqrt{1-k} = \sqrt{1-4.24520 \times 10^{-6}} = 0.99999787740$$

Confirmation: Schwarzschild gravitational time dilation at R_1 :

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{2GM}{R_1 c^2}} = \sqrt{1 - \frac{2.65437 \times 10^{20}}{6.25264 \times 10^{25}}} = \sqrt{1 - 4.24520 \times 10^{-6}} = 0.99999787740$$

Under static mass, $DTD_1/DTD_2 = \sqrt{R_2/R_1}$: time dilation depends only on radii (Table 3, column $M = M$).

Case 2: Static Radius

When $R_1 = R_2$ and $M_1 > M_2$:

$$k = \underbrace{\left(\frac{I_2}{I_1} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2 R_2^2}{M_1 R_1^2} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2}{M_1} \right)^{1/L}}_{L=3}$$

$$k = \left(\frac{M_2}{M_1} \right)^{1/3} \quad (39)$$

Example (Case 2): S_1 : $M_1 = 1 \times 10^{30}$ kg, S_2 : $M_2 = 1 \times 10^{24}$ kg, both at $R = 1 \times 10^8$ m. Starting from the isometric equivalence with static radius ($R_1 = R_2$, $L = 3$):

$$k = \left(\frac{M_2}{M_1} \right)^{1/3} = \left(\frac{1 \times 10^{24}}{1 \times 10^{30}} \right)^{1/3} = (1 \times 10^{-6})^{1/3} = 1 \times 10^{-2}$$

$$= 1.00000 \times 10^{-2}$$

Since $k = (k_2/k_1)^{1/3}$ where $k_i = D_i/D_{crit}$ and $D = M/R$:

$$k_1 = \frac{M_1/R}{D_{crit}} = \frac{1 \times 10^{30}/1 \times 10^8}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-5}$$

$$k_2 = k_1 \cdot k^3 = 1.48523 \times 10^{-5} \times (1.00000 \times 10^{-2})^3$$

$$= 1.48523 \times 10^{-5} \times 1.00000 \times 10^{-6} = 1.48523 \times 10^{-11}$$

$$\frac{d\tau}{dt} = \frac{\sqrt{1-k_1}}{\sqrt{1-k_2}} = \frac{\sqrt{1-1.48523 \times 10^{-5}}}{\sqrt{1-1.48523 \times 10^{-11}}}$$

$$= 9.9999257381 \times 10^{-1}$$

Confirmation: Schwarzschild gravitational time dilation:

$$\frac{d\tau}{dt} = \frac{\sqrt{1 - \frac{2GM_1}{Rc^2}}}{\sqrt{1 - \frac{2GM_2}{Rc^2}}}$$

$$= \frac{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^8)(8.98755 \times 10^{16})}}}{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{24})}{(1 \times 10^8)(8.98755 \times 10^{16})}}}$$

$$= \frac{\sqrt{1 - 1.48523 \times 10^{-5}}}{\sqrt{1 - 1.48523 \times 10^{-11}}} = 9.9999257381 \times 10^{-1}$$

Under static radius, $DTD_1/DTD_2 = \sqrt{M_1/M_2}$: time dilation depends only on mass (Table 3, column $R = R$).

Case 3: Constant Density

When $M \propto R^3$ and $R_1 > R_2$:

$$k = \underbrace{\left(\frac{I_2}{I_1} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2 R_2^2}{M_1 R_1^2} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{R_2^3 R_2^2}{R_1^3 R_1^2} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{R_2^5}{R_1^5} \right)^{1/L}}_{L=5}$$

$$k = \left(\frac{R_2^5}{R_1^5} \right)^{1/5} = \frac{R_2}{R_1} \quad (40)$$

Example (Case 3): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^8$ m. S_2 : $M_2 = 1 \times 10^{27}$ kg, $R_2 = 1 \times 10^7$ m. Both at $\rho = 2.38732 \times 10^5$ kg/m³. Starting from the isometric equivalence ($M \propto R^3$, $L = 5$):

$$k = \frac{R_2}{R_1} = \frac{1 \times 10^7}{1 \times 10^8} = 1 \times 10^{-1} = 1.00000 \times 10^{-1}$$

Since $D = M/R \propto R^2$ at constant density, $k_2/k_1 = (R_2/R_1)^2 = k^2$ where $k_i = D_i/D_{crit}$:

$$k_1 = \frac{M_1/R_1}{D_{crit}} = \frac{1 \times 10^{30}/1 \times 10^8}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-5}$$

$$k_2 = k_1 \cdot k^2 = 1.48523 \times 10^{-5} \times (1.00000 \times 10^{-1})^2 = 1.48523 \times 10^{-5} \times 1.00000 \times 10^{-2} = 1.48523 \times 10^{-7}$$

$$\frac{d\tau}{dt} = \frac{\sqrt{1-k_1}}{\sqrt{1-k_2}} = \frac{\sqrt{1-1.48523 \times 10^{-5}}}{\sqrt{1-1.48523 \times 10^{-7}}} = 9.9999264807 \times 10^{-1}$$

Confirmation: Schwarzschild gravitational time dilation:

$$\begin{aligned} \frac{d\tau}{dt} &= \frac{\sqrt{1 - \frac{2GM_1}{R_1 c^2}}}{\sqrt{1 - \frac{2GM_2}{R_2 c^2}}} \\ &= \frac{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^8)(8.98755 \times 10^{16})}}}{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{27})}{(1 \times 10^7)(8.98755 \times 10^{16})}}} \\ &= \frac{\sqrt{1 - 1.48523 \times 10^{-5}}}{\sqrt{1 - 1.48523 \times 10^{-7}}} = 9.9999264807 \times 10^{-1} \end{aligned}$$

This is the only constraint under which $(I_1/I_2)^{1/5} = \text{DTD}_1/\text{DTD}_2$; the density correction $(\rho_1/\rho_2)^{3/10}$ in Equation 23 is unity (Table 3, column $\rho = \rho$).

Case 4: Constant Inertia

When $M_1 R_1^2 = M_2 R_2^2$:

$$\begin{aligned} k &= \underbrace{\left(\frac{I_2}{I_1}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/L}}_{L=5} \\ k &= \left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/5} = (1)^{1/5} = 1 \end{aligned} \quad (41)$$

Example (Case 4): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^8$ m. S_2 : $M_2 = 1 \times 10^{32}$ kg, $R_2 = 1 \times 10^7$ m. Both have $I = MR^2 = 1 \times 10^{46}$ kg·m². Starting from the isometric equivalence:

$$I_1 = M_1 R_1^2 = (1 \times 10^{30})(1 \times 10^8)^2 = 1 \times 10^{46}$$

$$I_2 = M_2 R_2^2 = (1 \times 10^{32})(1 \times 10^7)^2 = 1 \times 10^{46}$$

$$k = \left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/5} = \left(\frac{1 \times 10^{46}}{1 \times 10^{46}}\right)^{1/5} = 1$$

Confirmation: Computing each system's D_{norm} independently:

$$k_1 = \frac{D_1}{D_{crit}} = \frac{M_1/R_1}{D_{crit}} = \frac{1 \times 10^{30}/1 \times 10^8}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-5}$$

$$k_2 = \frac{D_2}{D_{crit}} = \frac{M_2/R_2}{D_{crit}} = \frac{1 \times 10^{32}/1 \times 10^7}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-2}$$

$$\text{DTD}_1 = \sqrt{k_1} = 3.85387 \times 10^{-3}$$

$$\text{DTD}_2 = \sqrt{k_2} = 1.21870 \times 10^{-1}$$

$$\frac{\text{DTD}_2}{\text{DTD}_1} = 3.16228 \times 10^1$$

$$\left(\frac{\text{DTD}_2}{\text{DTD}_1}\right)^2 \cdot \left(\frac{R_2}{R_1}\right)^3 = (3.16228 \times 10^1)^2 \cdot \left(\frac{1 \times 10^7}{1 \times 10^8}\right)^3 = 1.00000 \times 10^3 \times 1.00000 \times 10^{-3} = 1$$

Equal inertia does not imply equal compactness: here $(I_2/I_1)^{1/5} = 1$ while the DTD ratio is 31.62, confirming Equation 35 (Table 3, column $I = I$).

General Case

When $M_1 \neq M_2$, $R_1 \neq R_2$, $M_1 \neq R_1^3$, and $M_2 \neq R_2^3$:

$$\begin{aligned} k &= \underbrace{\left(\frac{I_2}{I_1}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2}{M_1}\right)^{1/L}}_{L=5} \cdot \underbrace{\left(\frac{R_2}{R_1}\right)^{2/L}}_{L=5} \\ k &= \left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/5} = \left(\frac{M_2}{M_1}\right)^{1/5} \cdot \left(\frac{R_2}{R_1}\right)^{2/5} \end{aligned} \quad (42)$$

Example (General Case): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^7$ m. S_2 : $M_2 = 1 \times 10^{20}$ kg, $R_2 = 2 \times 10^{-7}$ m. Starting from the isometric equivalence:

$$\begin{aligned} k &= \left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/5} = \left(\frac{(1 \times 10^{20}) \cdot (2 \times 10^{-7})^2}{(1 \times 10^{30}) \cdot (1 \times 10^7)^2}\right)^{1/5} \\ &= \left(\frac{4.00000 \times 10^6}{1 \times 10^{44}}\right)^{1/5} = (4.00000 \times 10^{-38})^{1/5} \\ &= 3.31445 \times 10^{-8} \end{aligned}$$

Since $k_i = D_i/D_{crit}$ and $D = M/R$:

$$k_1 = \frac{M_1/R_1}{D_{crit}} = \frac{1 \times 10^{30}/1 \times 10^7}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-4}$$

$$k_2 = \frac{M_2/R_2}{D_{crit}} = \frac{1 \times 10^{20}/2 \times 10^{-7}}{6.73295 \times 10^{26}} = 7.42617 \times 10^{-1}$$

$$\begin{aligned}\frac{d\tau}{dt} &= \frac{\sqrt{1-k_2}}{\sqrt{1-k_1}} = \frac{\sqrt{1-7.42617 \times 10^{-1}}}{\sqrt{1-1.48523 \times 10^{-4}}} \\ &= 5.07368 \times 10^{-1}\end{aligned}$$

Confirmation: Schwarzschild gravitational time dilation:

$$\begin{aligned}\frac{d\tau}{dt} &= \frac{\sqrt{1 - \frac{2GM_2}{R_2 c^2}}}{\sqrt{1 - \frac{2GM_1}{R_1 c^2}}} \\ &= \frac{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(2 \times 10^{-7})(8.98755 \times 10^{16})}}}{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^7)(8.98755 \times 10^{16})}}} \\ &= \frac{\sqrt{1 - 7.42617 \times 10^{-1}}}{\sqrt{1 - 1.48523 \times 10^{-4}}} \\ &= 5.07368 \times 10^{-1}\end{aligned}$$

All identities from Section 5.5 hold without constraint; numerical verification appears in Table 3, column General.

Example (General Case — Comparison of Relative Time Paces): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^7$ m. S_2 : $M_2 = 1 \times 10^{20}$ kg, $R_2 = 2 \times 10^{-7}$ m.

Linear scale factor (Equation 17):

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{(1 \times 10^{30})(2 \times 10^{-7})}{(1 \times 10^{20})(1 \times 10^7)} = 2.00000 \times 10^{-4}$$

$$D_{norm} = \frac{M_2/R_2}{D_{crit}} = \frac{1 \times 10^{20}/2 \times 10^{-7}}{6.73295 \times 10^{26}} = 7.42617 \times 10^{-1}$$

$$\frac{d\tau}{dt} = \sqrt{1 - D_{norm}} = \sqrt{1 - 7.42617 \times 10^{-1}} = 5.07330 \times 10^{-1}$$

Confirmation: Schwarzschild gravitational time dilation for each system:

S_1 :

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^7)(8.98755 \times 10^{16})}} = \sqrt{1 - 1.48523 \times 10^{-4}} = 9.99926 \times 10^{-1}$$

S_2 :

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{20})}{(2 \times 10^{-7})(8.98755 \times 10^{16})}} = \sqrt{1 - 7.42617 \times 10^{-1}} = 5.07330 \times 10^{-1}$$

Relative pace of time:

$$\frac{\sqrt{1-k_2}}{\sqrt{1-k_1}} = \frac{5.07330 \times 10^{-1}}{9.99926 \times 10^{-1}} = 5.07368 \times 10^{-1}$$

Appendix B Pair Comparison Tables

Tables 2 and 3 present the complete system properties and derived ratios for six system pairs spanning the constraint cases from the derivations in Appendix A: static mass ($M=M$), static radius ($R=R$), constant density ($\rho = \rho$), constant inertia ($I=I$), an unconstrained general case, and a small-scale STE example. The isometric equivalence scale factor $(I_1/I_2)^{1/5}$ equals the DTD ratio only under constant density (Case 3 and the STE Example); in all other cases the universal STE (Equation 23) with its density correction is required. The inertia recovery identity $(DTD_1/DTD_2)^2(R_1/R_2)^3 = I_1/I_2$ holds in every case. The table is thus the numerical demonstration both of the original fifth-root inertia hypothesis (exact in the isometric column) and of its corrected universal form (exact in every column).

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all $\LaTeX$ source files, derivation documents, and revision history, are publicly available at: https://github.com/denverdata/academic_InertialRelativity-deriving-gr-sr-from-ste.

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Table 2 System properties for each pair comparison case.

	Case 1 ($M=M$)	Case 2 ($R=R$)	Case 3 ($\rho=\rho$)	Case 4 ($l=l$)	General	STE Example
m_1 (kg)	2.00000000e+30	1.00000000e+30	1.000000000000000000000000000000e+30	1.00000000e+32	1.00000000e+20	8.000000000000000000000000000000e+0
m_2 (kg)	2.00000000e+30	1.00000000e+24	1.000000000000000000000000000000e+27	1.00000000e+30	1.00000000e+30	1.000000000000000000000000000000e+0
r_1 (m)	3.00000000e+3	1.00000000e+8	1.000000000000000000000000000000e+8	1.00000000e+7	2.00000000e-7	2.000000000000000000000000000000e+0
r_2 (m)	7.00000000e+8	1.00000000e+8	1.000000000000000000000000000000e+7	1.00000000e+8	1.00000000e+7	1.000000000000000000000000000000e+0
ρ_1 (kg/m ³)	1.768388257e+19	2.387324146e+5	2.38732414644666471556605962123e+5	2.387324146e+10	2.984155183e+39	2.38732414644666471556605962123e-1
ρ_2 (kg/m ³)	1.392025741e+3	2.387324146e-1	2.38732414644666471556605962123e+5	2.387324146e+5	2.387324146e+8	2.38732414644666471556605962123e-1
I_1 (kg.m ²)	7.20000000e+36	4.00000000e+45	4.000000000000000000000000000000e+45	4.00000000e+45	1.60000000e+6	1.280000000000000000000000000000e+1
I_2 (kg.m ²)	3.92000000e+47	4.00000000e+39	4.000000000000000000000000000000e+40	4.00000000e+45	4.00000000e+43	4.000000000000000000000000000000e-1
D_1 (kg/m)	6.66666667e+26	1.00000000e+22	1.000000000000000000000000000000e+22	1.00000000e+25	5.00000000e+26	4.000000000000000000000000000000e+0
D_2 (kg/m)	2.857142857e+21	1.00000000e+16	1.000000000000000000000000000000e+20	1.00000000e+22	1.00000000e+23	1.000000000000000000000000000000e+0
D_{1_norm}	9.901553801e-1	1.485233070e-5	1.48523307019954106298130834181e-5	1.485233070e-2	7.426165351e-1	5.9409322807981642519252336725e-27
D_{2_norm}	4.243523058e-6	1.485233070e-11	1.48523307019954106298130834181e-7	1.485233070e-5	1.485233070e-4	1.48523307019954106298130834181e-27
r_{s_1} (m)	2.970464108e+3	1.485232054e+3	1.48523205382373310037190573048e+3	1.485232054e+5	1.485232054e-7	1.18818564305898648029752458438e-26
r_{s_2} (m)	2.970464108e+3	1.485232054e-3	1.48523205382373310037190573048e+0	1.485232054e+3	1.485232054e+3	1.48523205382373310037190573048e-27
gtd_1	9.922347228e-2	9.999925738e-1	9.99992573812156748392869162331e-1	9.925460591e-1	5.073302407e-1	9.999999999999999999999999997030e-1
gtd_2	9.999978782e-1	1.00000000e+0	9.99999925738394551420323081145e-1	9.999925738e-1	9.999257356e-1	9.99999999999999999999999999257e-1
$dt d_1$	9.950655155e-1	3.853872170e-3	3.85387216990852080205409550136e-3	1.218701387e-1	8.617520149e-1	7.70774433981704160410819100273e-14
$dt d_2$	2.059981325e-3	3.853872170e-6	3.85387216990852080205409550136e-4	3.853872170e-3	1.218701387e-2	3.85387216990852080205409550136e-14

Table 3 Derived ratios across constraint cases.

	Case 1 ($M=M$)	Case 2 ($R=R$)	Case 3 ($\rho=\rho$)	Case 4 ($l=l$)	General	STE Example
$dt d_1 / dt d_2$	4.830458915e+2	1.00000000e+3	1.000000000000000000000000000000e+1	3.162277660e+1	7.071067812e+1	2.000000000000000000000000000000e+0
$\sqrt{D_1/D_2}$	4.830458915e+2	1.00000000e+3	1.000000000000000000000000000000e+1	3.162277660e+1	7.071067812e+1	2.000000000000000000000000000000e+0
$\sqrt{(m_1 r_2)/(m_2 r_1)}$	4.830458915e+2	1.00000000e+3	1.000000000000000000000000000000e+1	3.162277660e+1	7.071067812e+1	2.000000000000000000000000000000e+0
$\sqrt{(I_1/I_2)(r_2/r_1)^3}$	4.830458915e+2	1.00000000e+3	1.000000000000000000000000000000e+1	3.162277660e+1	7.071067812e+1	2.000000000000000000000000000000e+0
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.830458915e+2	1.00000000e+3	1.000000000000000000000000000000e+1	3.162277660e+1	7.071067812e+1	2.000000000000000000000000000000e+0
$(m_1 r_2)/(m_2 r_1)$	2.333333333e+5	1.00000000e+6	1.000000000000000000000000000000e+2	1.00000000e+3	5.00000000e+3	4.000000000000000000000000000000e+0
D_1/D_2	2.333333333e+5	1.00000000e+6	1.000000000000000000000000000000e+2	1.00000000e+3	5.00000000e+3	4.000000000000000000000000000000e+0
$(I_1/I_2)(r_2/r_1)^3$	2.333333333e+5	1.00000000e+6	1.000000000000000000000000000000e+2	1.00000000e+3	5.00000000e+3	4.000000000000000000000000000000e+0
gtd_1/gtd_2	9.922368281e-2	9.999925738e-1	9.99992648073216231086442988774e-1	9.925534300e-1	5.073679200e-1	9.999999999999999999999999997772e-1
$\sqrt{1-dtd_1^2}/\sqrt{1-dtd_2^2}$	9.922026831e-2	9.999925738e-1	9.99992648068185132354286661810e-1	9.925534249e-1	5.073674192e-1	9.999999999999999999999999997772e-1
I_1/I_2	1.836734694e-11	1.00000000e+6	1.000000000000000000000000000000e+5	1.00000000e+0	4.00000000e-38	3.200000000000000000000000000000e+1
$(I_1/I_2)^{1/2}$	4.285714286e-6	1.00000000e+3	3.16227766016837933199889354443e+2	1.00000000e+0	2.00000000e-19	5.65685424949238019520675489684e+0
$(\rho_1/\rho_2)^{1/5}(r_1/r_2)$	7.125400598e-3	1.584893192e+1	1.000000000000000000000000000000e+1	1.00000000e+0	3.314454017e-8	2.000000000000000000000000000000e+0
$(I_1/I_2)^{1/5}$	7.125400598e-3	1.584893192e+1	1.000000000000000000000000000000e+1	1.00000000e+0	3.314454017e-8	2.000000000000000000000000000000e+0
$(dtd_1/dtd_2)^2(r_1/r_2)^3$	1.836734694e-11	1.00000000e+6	1.000000000000000000000000000000e+5	1.00000000e+0	4.00000000e-38	3.200000000000000000000000000000e+1
m_1/m_2	1.00000000e+0	1.00000000e+6	1.000000000000000000000000000000e+3	1.00000000e+2	1.00000000e-10	8.000000000000000000000000000000e+0
$\sqrt{m_1/m_2}$	1.00000000e+0	1.00000000e+3	3.16227766016837933199889354443e+1	1.00000000e+1	1.00000000e-5	2.82842712474619009760337744842e+0
r_2/r_1	2.333333333e+5	1.00000000e+0	1.000000000000000000000000000000e-1	1.00000000e+1	5.00000000e+13	5.000000000000000000000000000000e-1
$\sqrt{r_2/r_1}$	4.830458915e+2	1.00000000e+0	3.16227766016837933199889354443e-1	3.162277660e+0	7.071067812e+6	7.07106781186547524400844362105e-1